GST Billing, Inventory & Technical Quotation System

Complete Architecture, Feature Walkthrough & Client Presentation Guide

Key Value Proposition to Pitch to Client

This software is an all-in-one, offline-first GST Billing, Inventory Management, and Technical Quotation system. All data stays 100% private and secure on the client's own machine. No recurring subscription fees, no internet dependency for daily billing, instant auto-tax calculations, automatic stock deduction, and 1-click Quotation-to-Tax Invoice conversion.

1. Executive Overview & System Architecture

The platform is built using modern industry standards: a React 19 + Vite frontend combined with a local Node.js Express server and a dual-storage engine (Embedded SQLite database with WAL-mode + JSON backups). It provides desktop-grade speed with zero cloud downtime risk.

- Frontend: React 19, Vite 7, Responsive layout, Offline PWA with instant UI updates.
- Backend API: Express.js 5 running on localhost port 47371 with strict local security policies.
- Database Engine: Embedded SQLite (better-sqlite3) with ACID transactions and structured SQL schema for master catalogues, stock balances, customers, and quotations.
- Data Ownership: 100% offline data storage. Invoices and financial figures never leave the client premises unless they explicitly export or sync to their personal Google Drive.

2. Dashboard & Real-Time Business Control

The Dashboard is the operational command center designed for business owners, accountants, and cashiers to monitor cash flow and pending collections in real time.

Key Highlights to Demonstrate:

- Financial KPI Cards: Real-time counters for Total Sales Revenue, Received Payments, and Outstanding Balances.
- Multi-Currency Support: Automatic currency detection (INR, USD, EUR, GBP, AED, etc.) with converted summaries.
- Live Invoice Ledger: Comprehensive data table with instant search (by bill number, client name, GSTIN) and filter tabs (All, Paid, Unpaid, Overdue, Cancelled).
- Quick Row Actions: One-click buttons to View/Print PDF, Edit Bill, Duplicate Invoice, Cancel Bill, and Record Payments.
- Smart Notification Center: Bell icon alerting low-stock items, overdue payment deadlines, and GST filing schedules.

3. Billing & Invoicing Engine (Core Demo Feature)

A versatile GST invoicing engine capable of generating all mandatory Indian tax documents with automatic tax calculations.

Supported Document Types:

- Tax Invoice: Standard GST B2B / B2C tax invoice with CGST + SGST (intra-state) or IGST (inter-state).
- Quotation (QUO): Priced quotation for works or supply. Fully editable and converts to a Tax Invoice in one click once client approves.
- Proforma Invoice: Formal estimation for advance payment requests prior to actual goods dispatch.
- Delivery Challan: For moving goods between sites/warehouses without creating a taxable sale.
- Credit Note: For sales returns, damaged goods, or post-sale discounts (automatically refunds stock).
- Bill of Supply: For composition taxpayers or supply of GST-exempted goods.

Billing Capabilities to Highlight:

- Automatic GST Logic: Matches Supplier State with Client State; automatically splits 18% into 9% CGST + 9% SGST for local sales, or charges 18% IGST for interstate.

- Order & Project Details: Dedicated fields for Buyer Purchase Order (PO) Number & Date, Work Scope / Project Details, Service Period (From/To), and Vehicle / Delivery Note numbers.

- Live Dynamic UPI QR Code: Invoice PDF prints a scannable dynamic UPI QR code containing the bill total and payee UPI ID for instant customer payment.

- Print Formats & Branding: Supports A4, Letter, Half-page, and 3-inch thermal POS receipt layouts. Displays Company Logo, Authorized Signature, and Official Stamp.

4. Inventory & Stock Management

A real-time stock control module that links directly into the billing engine for automated stock deduction and replenishment.

Key Inventory Features:

- Product Master Catalogue: Maintains Product Name, SKU, HSN/SAC code, Category, Unit (Pcs, Set, Kg, Box, Mtr, Job), and Tax Rate.

- Dual Pricing Mechanism: Stores both Purchase Price (what business paid the supplier) and Selling Price (billing rate for customers) to track profit margins.

- Live Stock Deduction: When a Tax Invoice is saved, the exact quantity of products sold is automatically subtracted from inventory balances.

- Low-Stock Alerts: Configurable threshold (e.g. 5 units). System raises automatic warnings when items run low or hit zero.

- Sales Return Restock: Issuing a Credit Note automatically credits the returned quantity back into stock.

- Excel & CSV Tools: Bulk upload and download products via CSV/Excel spreadsheets for easy initial setup or stock audits.

5. Technical Quotation & Cold-Room Sizing (Specialized)

A built-in engineering calculation and quotation module specially created for HVAC, refrigeration, and cold storage contractors.

Technical Workflow:

- Heat Load & Sizing Calculator: Takes Room Dimensions (Length, Width, Height), Application Type (Chiller +4°C, Freezer -20°C, Milk BMC), and Ambient Temperature.

- Equipment Recommendation: Automatically calculates required cooling capacity in kW / HP and suggests matching Emerson / Daikin condensing units and evaporators.

- Full BOQ Quotation: Assembles equipment, piping, refrigerant, accessories, electricals, and installation/commissioning charges.

- Site & Project Association: Associates the quotation with specific Customer ID and Project Site name.

- 1-Click Conversion: Once the customer approves the proposal, a single button click converts the entire engineering quotation into an official Tax Invoice.

6. Client & Customer Ledger Management

Centralized party management maintaining complete financial history, billing records, and outstanding ledgers.

- Client Directory: Stores Client Business Name, Contact Person, Phone, Email, Billing Address, State Code, and GSTIN.

- GSTIN Validation: Validates 15-digit GSTIN structure and auto-detects client state code to eliminate tax errors.

- Party Outstanding Ledger: Tracks total billed amount, total payments received, and net balance due per client.

- Customer Sites / Projects: Allows multiple project sites or delivery branch addresses under a single parent company.

7. Purchases, Expenses & Receipts

Complete inward accounting ensuring the client has accurate profit & loss figures and Input Tax Credit (ITC) tracking.

- Purchase Bills: Record supplier purchase invoices. Automatically increases inventory stock levels upon receiving goods.

- Expense Tracker: Log operational overheads (Rent, Electricity, Logistics, Site Travel, Tools, Labour).

- Receipt Vouchers: Issue official payment receipts when customers pay against pending invoices with payment mode details (NEFT, UPI, Cheque, Cash).

8. GST Compliance & Returns Portal

Eliminates CA dependency for basic monthly compliance by generating portal-ready export files.

- GSTR-1 JSON Export: Generates GST Portal compliant JSON files for outward B2B invoices, B2CS, and Credit Notes for direct upload to gst.gov.in.

- GSTR-3B Summary: Pre-aggregates total taxable turnover, integrated tax, central tax, state tax, and cess values.

- GSTR-2B Reconciliation: Reconciles purchase records against portal 2B data to claim maximum legitimate Input Tax Credit.

- TDS & TCS Reports: Computes tax deducted at source under Section 194Q and tax collected at source under Section 206C.

9. Data Security, Privacy & Backup Systems

Enterprise-level privacy controls ensuring complete business confidentiality.

- Local Security PIN: Optional App Lock PIN preventing unauthorized staff from viewing financial books on the counter PC.

- Automated Daily Snapshots: System creates automatic timestamped backups inside the local data folder every day.

- One-Click Backup & Restore: Export the entire company database into a compressed ZIP file to move to another laptop or keep on a USB drive.

- Optional Google Drive Sync: Direct encrypted backup upload to the user's own private Google Drive storage.

10. Step-by-Step Client Demo Script (Recommended Flow)

Follow this exact chronological sequence during the client presentation for the strongest impact:

- Step 1: Dashboard Tour: Show the clean dashboard, financial cards, and explain that all data is running 100% locally on this computer without internet dependency.

- Step 2: Inventory Overview: Open Products tab. Show how items have both Purchase Price, Selling Price, and current Stock count.

- Step 3: Create Quotation: Open New Invoice. Select Document Type = "Quotation". Select client, add project work details, pick products. Show live PDF preview with logo and bank details.

- Step 4: Convert to Tax Invoice: Show how client approval turns the Quotation into a Tax Invoice in 1 click. Save the invoice.

- Step 5: Verify Stock Cut: Go back to Products tab and prove that the stock was automatically deducted! Show the low-stock alert.

- Step 6: Payment & Receipt: Record a partial payment on the invoice and generate an instant payment receipt voucher.

- Step 7: GST Return Export: Open GST Returns tab and show the ready-to-upload GSTR-1 JSON export.